

SPRING NOTES 2024

SQL Fundamentals

Database Design · DDL & DML Commands · Constraints · Keys & Queries

[Database Design](#)[Keys & Constraints](#)[DDL Commands](#)[DML Commands](#)[SELECT Queries](#)



SECTION 01

The Problem — Data Redundancy

When all student data is stored in a single flat table, the same student's details (name, phone, email) are repeated for every course or month entry. This is called **data redundancy**.

stno	stname	phno	email	coursename	month	fees
1	abc	123	abc@gmail.com	Java	Jan	12000
1	abc	123	abc@gmail.com	Java	Feb	12000
2	bcd	234	bcd@gmail.com	SQL	Jan	12000

⚠ **Problem:** Student "abc" appears twice — once for Jan and once for Feb. Name, phone and email are stored repeatedly. Any update must be done in multiple rows, risking inconsistency.

✅ **Solution — Normalization:** Split into three related tables: student, course, and fees. Link them using Primary & Foreign Keys so each piece of data is stored only once.



SECTION 02

Keys — Primary Key & Foreign Key



Primary Key (PK)

Uniquely identifies every record. Cannot be **NULL**. No duplicates allowed. Only *one* PK per table.

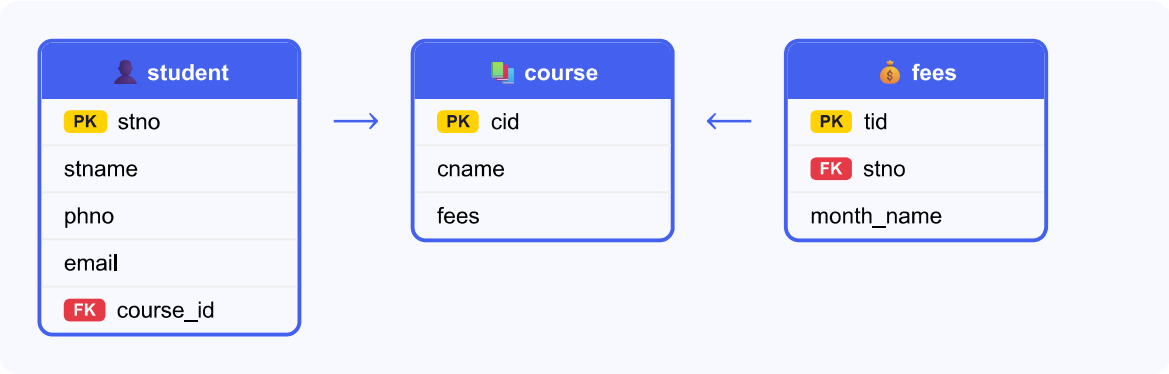


Foreign Key (FK)

References the PK of another table. Must have the **same data type & size** as the PK, but can have a different name.



🔴 **Order of creation:** Always create the parent table (with PK) before the child table (with FK).
Correct order: course → student → fees



SECTION 03

SQL Command Categories

DDL

CREATE
ALTER
DROP
TRUNCATE

DML

INSERT
UPDATE
DELETE

DQL

SELECT

TCL

COMMIT
ROLLBACK
SAVEPOINT

Category	Full Form	Purpose	Auto-Commit
DDL	Data Definition Language	Define / modify table structure	✔ Yes
DML	Data Manipulation Language	Insert, update, delete data	✘ Manual
DQL	Data Query Language	Read / retrieve data	N/A
TCL	Transaction Control Language	Manage transactions	Manual



SECTION 04

DDL — CREATE TABLE

```
/* Step 1: Create the database */
CREATE DATABASE stdb;

/* Step 2: Open the database */
USE stdb;

/* Step 3: Create COURSE table first (parent - no FK) */
CREATE TABLE course (
    cid      INT(3)          PRIMARY KEY,
    cname    VARCHAR(20),
    fees     INT(5)
);

/* Step 4: Create STUDENT table with FK to course */
CREATE TABLE student (
    stno      INT(3)          PRIMARY KEY,
    stname    VARCHAR(50),
    phno      VARCHAR(10),
    email     VARCHAR(50),
    course_id INT(3),
    CONSTRAINT course_fk FOREIGN KEY(course_id)
        REFERENCES course(cid)
);

/* Step 5: Create FEES table with FK to student */
CREATE TABLE fees (
    tid       INT(3)          PRIMARY KEY,
    stno      INT(3),
    month_name VARCHAR(10),
    CONSTRAINT student_fk FOREIGN KEY(stno)
        REFERENCES student(stno)
);
```



SECTION 05

DML — INSERT Records

```
/* Insert into course */
INSERT INTO course(cid, cname, fees) VALUES
    (1, 'Java', 12000),
    (2, 'SQL', 7000),
    (3, 'AI', 12000);

/* Insert students - all columns */
INSERT INTO student(stno, stname, phno, email, course_id) VALUES
    (2, 'Abi', '1234567890', 'Abi@cognizant.com', 1),
    (7, 'Dhanu Ganesh', '9237567890', 'Dhanu@cognizant.com', 3);

/* Insert students - skip optional columns */
INSERT INTO student(stno, stname, email, course_id) VALUES
    (5, 'Jay', 'Jay@cognizant.com', 1),
    (3, 'ABC', 'abc@cognizant.com', 1);

INSERT INTO student(stno, stname, phno, course_id) VALUES
    (1, 'Prajan', '9834567123', 2),
    (4, 'Priya', '9534567123', 2);

INSERT INTO student(stno, stname, course_id) VALUES (6, 'Lalith', 3);

/* Insert fees */
INSERT INTO fees VALUES
    (1,1,'Jan'), (2,2,'Jan'), (3,6,'Jan'), (4,1,'Feb');
```



SECTION 06
DML — UPDATE & DELETE

```
/* Disable safe update mode */
SET sql_safe_updates = 0;

/* Change name 'Abi' to 'Archana' */
UPDATE student
SET   stname = 'Archana'
WHERE stname = 'Abi';

/* Delete a specific row */
DELETE FROM student WHERE stno = 3;
```

⚠ **Always use WHERE** with UPDATE and DELETE. Without it, ALL rows will be affected.

Command	Removes	Can Rollback?	Resets Auto-Increment?
DELETE	Specific or all rows	✔ Yes	✘ No
TRUNCATE	All rows at once	✘ No	✔ Yes

DROP	Entire table	✘ No	N/A
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SECTION 07

DDL — ALTER TABLE

```
/* Change phno from VARCHAR to BIGINT */  
ALTER TABLE student  
    CHANGE COLUMN phno phno BIGINT(10);  
  
/* Increase email size to 200 */  
ALTER TABLE student  
    CHANGE COLUMN email email VARCHAR(200);  
  
/* Add a new column */  
ALTER TABLE student ADD COLUMN address VARCHAR(255);  
  
/* Remove a column */  
ALTER TABLE student DROP COLUMN address;
```



SECTION 08

Constraints

Constraints are **rules applied to columns** to ensure data accuracy and integrity.

**PRIMARY KEY**

Uniquely identifies each record. Cannot be NULL. No duplicates. Only one per table.

**FOREIGN KEY**

References PK of another table. Enforces referential integrity between tables.

**UNIQUE**

No duplicate values allowed. But can accept NULL. Multiple UNIQUE constraints allowed per table.

**NOT NULL**

Column must always have a value. NULL is not permitted in this column.

Constraint	Allows NULL?	Allows Duplicates?	Per Table
PRIMARY KEY	✗ No	✗ No	Only 1
FOREIGN KEY	✓ Yes	✓ Yes	Multiple
UNIQUE	✓ Yes	✗ No	Multiple
NOT NULL	✗ No	✓ Yes	Multiple

```
CREATE TABLE student (
    stno    INT(3)      PRIMARY KEY,      -- unique + not null
    stname  VARCHAR(50) NOT NULL,         -- must always have value
    email   VARCHAR(100) UNIQUE,          -- no duplicate emails
    phno    VARCHAR(10),                  -- optional (can be null)
    cid     INT(3),
    CONSTRAINT fk_course FOREIGN KEY(cid) REFERENCES course(cid)
);
```



SECTION 09

DQL — SELECT Queries

```
/* Select all */
SELECT * FROM student;

/* Specific columns */
SELECT stno, stname, email FROM student;

/* Filter with WHERE */
SELECT * FROM student WHERE course_id = 1;

/* Sort results */
SELECT * FROM student ORDER BY stname ASC;

/* Join student + course */
SELECT s.stno, s.stname, c.cname, c.fees
FROM   student s
JOIN   course  c ON s.course_id = c.cid;
```


 QUICK REF

Data Types Cheat Sheet

Data Type	Description	Example
INT(n)	Whole numbers	stno INT(3)
BIGINT(n)	Large numbers (phone)	phno BIGINT(10)
VARCHAR(n)	Variable-length text	stname VARCHAR(50)
CHAR(n)	Fixed-length text	gender CHAR(1)
DECIMAL(p,s)	Exact decimals	price DECIMAL(8,2)
DATE	Date (YYYY-MM-DD)	dob DATE